26900144 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBCA301 - TIME SERIES ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Explain how time-series data differ from cross-sectional data. [U] / [CO1]
2. State the difference between short-term and long-term forecasting. [K] / [CO1]
3. Define the purpose of using mathematical transformation in time series data. [K] / [CO2]
4. List any two methods used for checking model adequacy. [K] / [CO3]
5. A dataset contains monthly sales and advertising expenditure. Identify whether it can be treated as a multivariate time series. [A] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Define a time series model and explain its purpose in time series analysis. [K] / [CO1]

7. Explain how time series models, autocorrelation, and partial autocorrelation work together in forecasting future observations. [U] / [CO1]

8. Examine the advantages and limitations of smoothing methods in time series analysis. [An] / [CO2]

9. Given a set of actual and forecasted values, compute the Mean Absolute Deviation (MAD) and Mean Absolute Percentage Error (MAPE). [A] / [CO2]

10. Explain the estimation of parameters in time-series regression models. [U] / [CO3]

11. Explain the structure and purpose of a Vector ARIMA model. [U] / [CO4]

12. Analyse how Bayesian methods differ from conventional forecasting approaches. [An] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Explain the importance of model evaluation and monitoring in time series forecasting. Discuss how graphical analysis and forecasting error measures can be used to assess the performance of a forecasting model. [U] / [CO2]

14. Explain the concept of statistical inference in the context of linear regression. Distinguish between point estimation, interval estimation, and hypothesis testing of regression parameters. [U] / [CO3]

15. Explain the principles of spectral analysis in time series. Analyse how frequency-domain analysis helps in identifying periodic patterns and cyclical behaviour. [U] / [CO4]

[15x2 = 30]